

CocoaPDF Strategic Corner Cases V1–V4

Fixture contract: this document intentionally combines common, adversarial, and future-scope cases. It is not a prose document; it is a deterministic test surface. Each section contains visible sentinel labels so conversion failures can be localized.

Canonical package name: CocoaPDF. The legacy `pdf2md` name may appear only as an import/CLI compatibility alias.

[[TOC source sentinel: a generated PDF may or may not create a real PDF outline/bookmark tree.]]

1. Plain Text, Paragraphs, Spacing, and Line Joining

SENTINEL-TEXT-001: A simple paragraph should survive as one Markdown paragraph with normal spaces, punctuation, and sentence boundaries.

SENTINEL-TEXT-002: This paragraph contains deliberately irregular spacing, a non-breaking space between Cocoa PDF, a narrow no-break space in 12345, and a thin space around A B. Normalize only where the converter policy explicitly allows it.

SENTINEL-TEXT-003: Soft line wrapping should be joined into one paragraph when the PDF visual line breaks are only layout artifacts. This sentence is intentionally long so that many PDF generators wrap it across several visual lines while the expected Markdown remains one paragraph.

SENTINEL-TEXT-004: Hard line breaks follow:
first hard line
second hard line
third hard line

SENTINEL-TEXT-005: Hyphenated wrap target words: micro-
service, co-
operate, re-
entry. Soft hyphen target: invisible soft hyphen.

SENTINEL-TEXT-006: Literal hyphenation that must not be repaired: state-of-the-art, end-to-end, mother-in-law, twenty-one, non-breaking hyphen A-B.

SENTINEL-TEXT-007: Dotted leader line for TOC-like detection: Introduction 7

2. Unicode, Encodings, Symbols, and Normalization

SENTINEL-UNICODE-001 Greek uppercase: Α Β Γ Δ Ε Ζ Η Θ Ι Κ Λ Μ Ν Ξ Ο Π Ρ Σ Τ Υ Φ Χ Ψ Ω.

SENTINEL-UNICODE-002 Greek lowercase: α β γ δ ε ζ η θ ι κ λ μ ν ξ ο π ρ σ τ υ φ χ ψ ω; variants: ς ϕ ϑ €.

SENTINEL-UNICODE-003 Latin diacritics composed: café naïve façade coöperate São Tomé Łódź Ćevapi Ångström smörgåsbord.

SENTINEL-UNICODE-004 Latin diacritics decomposed: café naïve façade coöperate São Tomé Ångström.

SENTINEL-UNICODE-005 Ligatures and compatibility: fi fl ffi ffl ff; plain equivalent: fi fl ffi ffl ff; Roman numerals: I II III IV V X.

SENTINEL-UNICODE-006 Math operators: ± × ÷ ≠ ≈ ≡ ≤ ≥ ∑ ∏ √ ∫ ∂ ∇ ∞ ∝ ∈ ∉ ∩ ∪ ⊆ ⊂ ⊕ ⊗ ∴ ∷.

SENTINEL-UNICODE-007 Arrows: ← ↑ → ↓ ↔ ↕ ⇒ ⇐ ⇨ ⇩ ⇧ ➡ ➠ ➞ ➟ ➡ ➠ ➞ ➟.

SENTINEL-UNICODE-008 Currency and legal symbols: \$ € £ ¥ ₹ ₪ ¢ ₩ ₮ ₦ ₧ © ® ™ § ¶ † ‡ №.

SENTINEL-UNICODE-009 Emoji single code points: 😊 😄 🚀 🍱 📄 🖋️ ✅ ❌ ⚠️.

SENTINEL-UNICODE-010 Emoji ZWJ and modifiers: 🇺🇸👤🏠👯🇺🇸 IN US.

SENTINEL-UNICODE-011 CJK: 中文测试 日本語テスト 한국어 테스트 漢字かなカナ.

SENTINEL-UNICODE-012 RTL visible text: العربية مرحبا بالعالم. עברית שלום עולם. Mixed LTR/RTL: ABC العربية 123 עברית XYZ.

SENTINEL-UNICODE-013 Indic and complex shaping: हिन्दी परीक्षण, বাংলা পরীক্ষা, தமிழ் சோதனை, తెలుగు పరీక్ష, ಕನ್ನಡ ಪರೀಕ್ಷೆ.

SENTINEL-UNICODE-014 Other scripts: Кириллица тест, Ελληνικά δοκιμή, Հայերեն փորձարկում, ქართული ტესტი, ไทยทดสอบ.

SENTINEL-UNICODE-015 Invisible/control-like characters named visibly: zero-width joiner [], zero-width non-joiner [], word joiner [], left-to-right mark [], right-to-left mark [].

3. Markdown Escaping and Literal Characters

SENTINEL-ESCAPE-001: Literal Markdown metacharacters in prose: backslash \ backtick ` asterisk * underscore _ braces {} brackets [] parentheses () hash # plus + minus - dot . exclamation ! pipe | angle &.

SENTINEL-ESCAPE-002: These should remain prose, not syntax: #noheading, -notlist, +notlist, *not emphasis*, *not emphasis*, 1.not ordered list.

SENTINEL-ESCAPE-003: Entity-looking text: © & © should not be double-decoded unexpectedly.

SENTINEL-ESCAPE-004: Literal HTML-looking text in code: `<div class="x">not active html</div>`.

4. Heading Hierarchy and Anchors

H1 Duplicate Title Sentinel

H2 Numbered and Plain

H3 1.2.3 Numbered Heading

H4 With Bold and *Italic* Inline Source

H5 ALL CAPS HEADING

H6 Small Heading

H2 Duplicate Title Sentinel

H2 Duplicate Title Sentinel

SENTINEL-ANCHOR-001: The explicit anchor target above should be available for internal link tests when the Markdown/PDF generator preserves anchors.

5. Inline Formatting

SENTINEL-INLINE-001: Plain, **bold**, *italic*, ***bold italic***, `inline_code()`, ~~strikethrough~~, underline via HTML fallback, highlight via HTML fallback.

SENTINEL-INLINE-002: Mixed styles inside one sentence: alpha **bravo** *charlie* **delta** echo `foxtrot_golf` hotel.

SENTINEL-INLINE-003: Superscript/subscript: H₂O, CO₂, x² + y² = z², E = mc².

SENTINEL-INLINE-004: Code span with punctuation: `a|b*c_d[0](x){y}`.

SENTINEL-INLINE-005: Backtick stress: `code span containing ` one backtick` and `code span containing `` two backticks`.

SENTINEL-INLINE-006: Faux-style text that should stay plain if source PDF does not encode style: BOLDLOOKINGCAPS italic-looking-words code-looking_word.

6. Links, Destinations, and Security

SENTINEL-LINK-001: External link: [CocoaPDF external HTTPS](#).

SENTINEL-LINK-002: Mail link: [Email maintainer](#).

SENTINEL-LINK-003: Telephone link: [Call placeholder](#).

SENTINEL-LINK-004: Internal link: [Jump to explicit anchor](#).

SENTINEL-LINK-005: URL with parentheses: [URL parentheses](#).

SENTINEL-LINK-006: Bare autolinks: <https://example.org/bare-url> and <mailto:bare@example.org>.

SENTINEL-LINK-SECURITY-001: Unsafe URI must be reported or rendered as plain text, not active: [unsafe javascript](#).

SENTINEL-LINK-SECURITY-002: Unsafe file URI must be reported or rendered as plain text, not active: [unsafe local file](#).

7. Lists and Nesting

SENTINEL-LIST-001 unordered list:

- alpha unordered item
- bravo unordered item with **bold** and `code`
- charlie unordered item that wraps in PDF because it contains a long explanation about list continuation, indentation, and visual line recovery across a layout-generated soft wrap.

SENTINEL-LIST-002 ordered list:

1. first ordered item
2. second ordered item
3. third ordered item

SENTINEL-LIST-003 ordered list starting at non-one:

7. item seven
8. item eight
9. item nine

SENTINEL-LIST-004 letter and roman markers source:

- a. letter item source
- b. second letter item source
- iv. roman item source
- v. second roman item source

SENTINEL-LIST-005 nested mixed list:

- 1. parent ordered alpha
 - child unordered alpha
 - child unordered bravo
 - 1. grandchild ordered one
 - 2. grandchild ordered two
- 2. parent ordered bravo
 - 1. child ordered one
 - deep bullet one
 - deep bullet two
 - 2. child ordered two

SENTINEL-LIST-006 task list source:

- ☒ checked task item
- ☐ unchecked task item
- ☒ uppercase checked task item

SENTINEL-LIST-007 singleton bullet that may be prose and should not be hallucinated as list without evidence:

- Singleton bullet-looking prose line.

SENTINEL-LIST-008 dash dialogue that should not become a list:

- This is dialogue with an em dash.
- This is another dialogue line.

8. Blockquotes

SENTINEL-QUOTE-001 simple quote line one.
simple quote line two with **bold** and `code`.

SENTINEL-QUOTE-002 quote with list:

- quoted bullet alpha
- quoted bullet bravo
- 1. quoted ordered alpha
- 2. quoted ordered bravo

SENTINEL-QUOTE-003 nested quote:

nested level two
nested level three

SENTINEL-QUOTE-004 quote with code:

```
def quoted_code(x):  
    return x + 1
```

9. Code Blocks

SENTINEL-CODE-001 fenced Python:

```
def alpha(value: int) -> str:  
    if value <= 0:  
        return "zero-or-negative"  
    return f"value={value}"
```

SENTINEL-CODE-002 blank lines, tabs, trailing-looking spaces:

```
line one  
  
    line with leading tab  
    line with four spaces  
pipe | backtick ` asterisk * underscore _ brackets [] braces {}
```

SENTINEL-CODE-003 long code line:

abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789__abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZUVWXYZ0123456789__END

SENTINEL-CODE-004 code block containing triple backticks:

```
```python
print("inner fenced block")
```
```

SENTINEL-CODE-005 HTML/XML code:

```
<section data-case="v1-4"><span>escaped & visible</span></section>
```

10. Horizontal Rules and Decorative Lines

SENTINEL-RULE-001 rule follows.

SENTINEL-RULE-002 text after rule. The line above is semantic thematic break, not a table border.

Heading underline adversarial case

SENTINEL-RULE-003: The underline above may be rendered as Setext heading by Markdown generators; converter must avoid misclassifying unrelated table borders as thematic breaks.

11. Tables — GFM, Alignment, Escapes, and Borderless-Like Patterns

SENTINEL-TABLE-001 simple GFM table:

Name	Qty	Price
cocoa	3	12.50
PDF beans	11	99.99
escaped pipe	1	a b

SENTINEL-TABLE-002 alignment table:

Left aligned	Center aligned	Right aligned
alpha	bravo	123
long wrapped cell text that may visually wrap in PDF	centered	456.78

SENTINEL-TABLE-003 numeric and decimal alignment:

Metric	2024	2025	Δ
Revenue	1,234.50	2,345.60	+1,111.10
Cost	-333.33	-444.44	-111.11
Ratio	0.125	0.875	+0.750

SENTINEL-TABLE-004 table-like prose that must not become a table:

Alpha Bravo Charlie
This is aligned prose, not a table, because it lacks repeated row semantics and it is a sentence-like paragraph.

SENTINEL-TABLE-005 key-value borderless table source:

Field	Value
Project	CocoaPDF
Scope	structured text-layer PDFs
Mode	Markdown-first with HTML fallback

12. Complex HTML Tables Requiring HTML Fallback

SENTINEL-HTML-TABLE-001 complex table:

Complex semantic table with spans, nested content, and code

Component	Evidence		Required output
	Geometry	Semantic prior	
Heading	font size, gap, bold ratio	tagged H1–H6 when available	Markdown ATX heading
Complex cell	- nested list inside cell - inline <code>code()</code> inside cell - Unicode αβγ and emoji 🍪		HTML table fallback
Rowspan example	shared geometry evidence	first semantic row	preserve rowspan
Rowspan continuation		second semantic row	do not flatten silently

13. Figures, Images, Captions, Floats, and SVG

SENTINEL-FIGURE-001 Markdown image with SVG data URI:

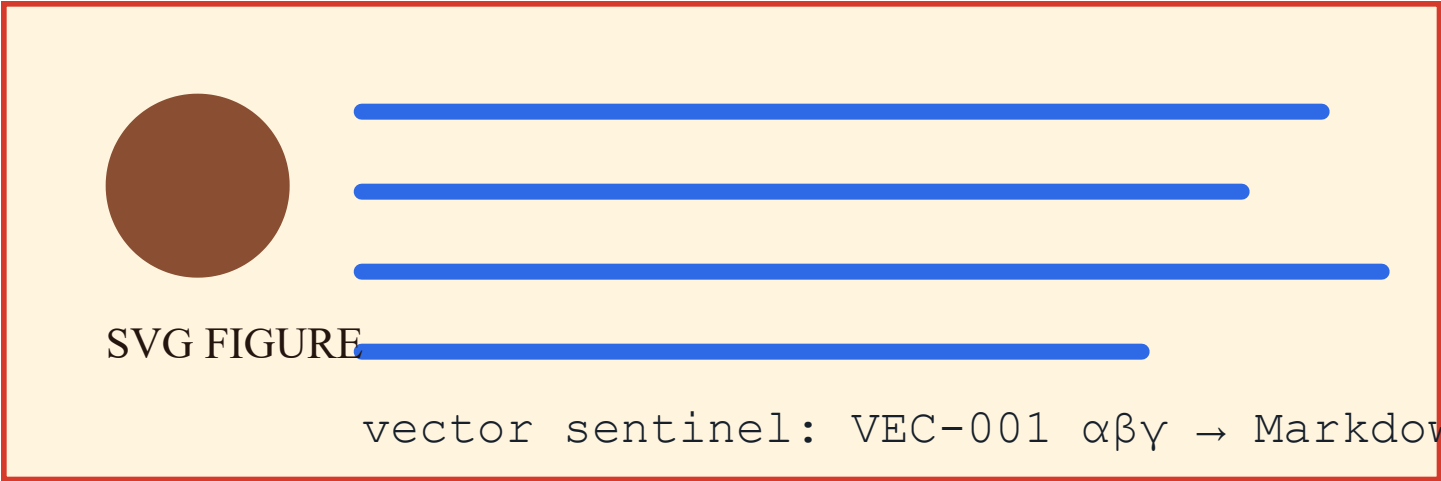


Figure 1. SVG vector figure with cocoa mark, blue lines, and visible vector sentinel text.

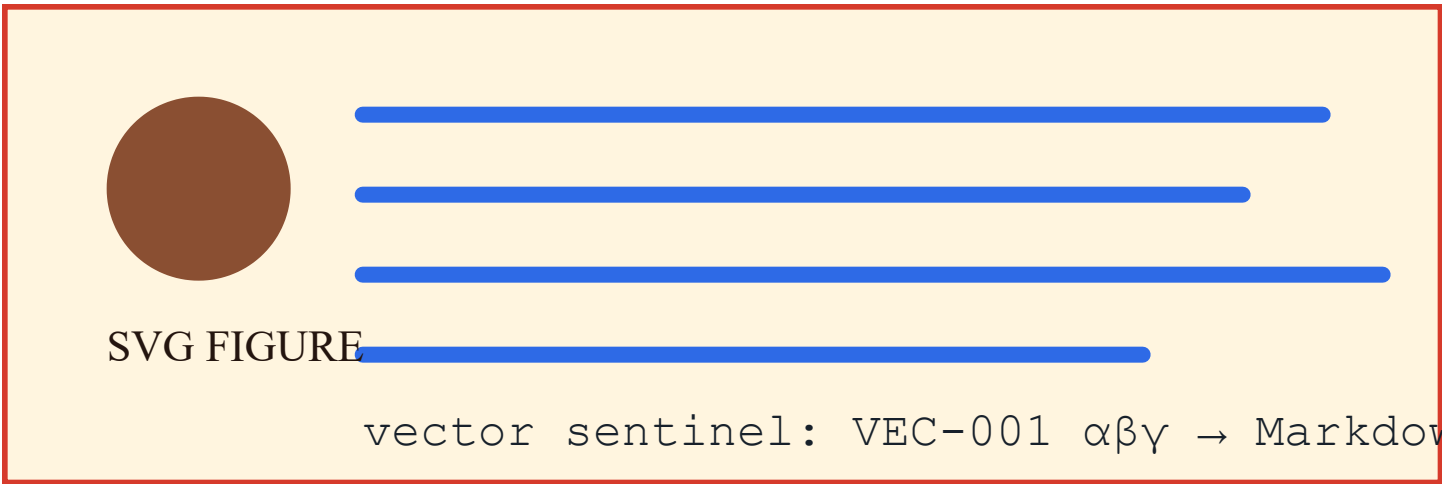
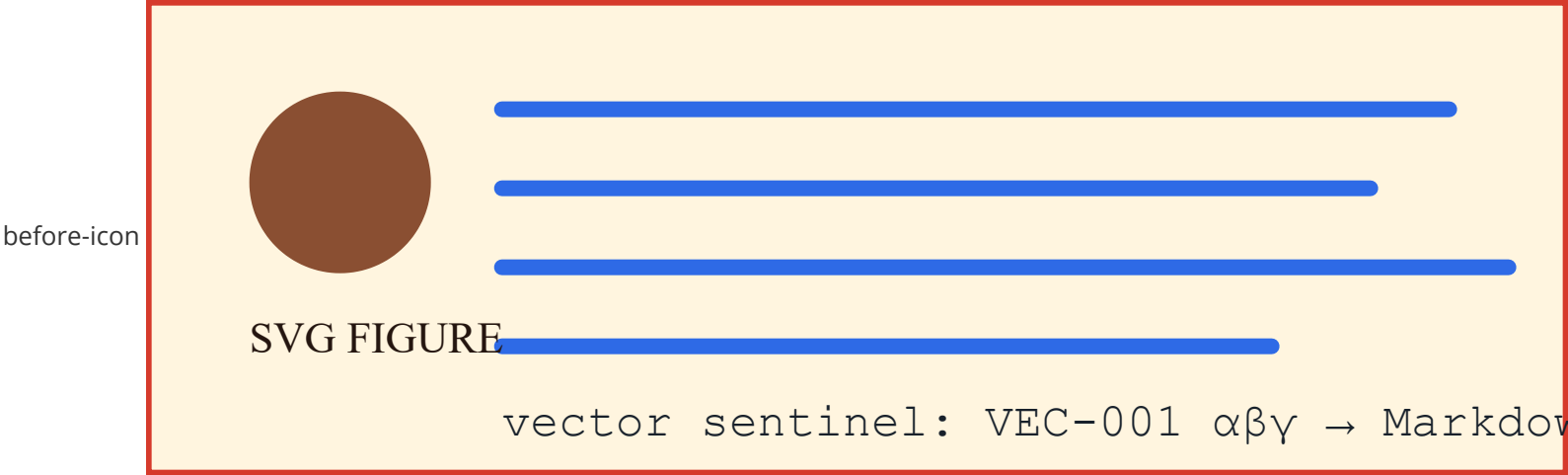


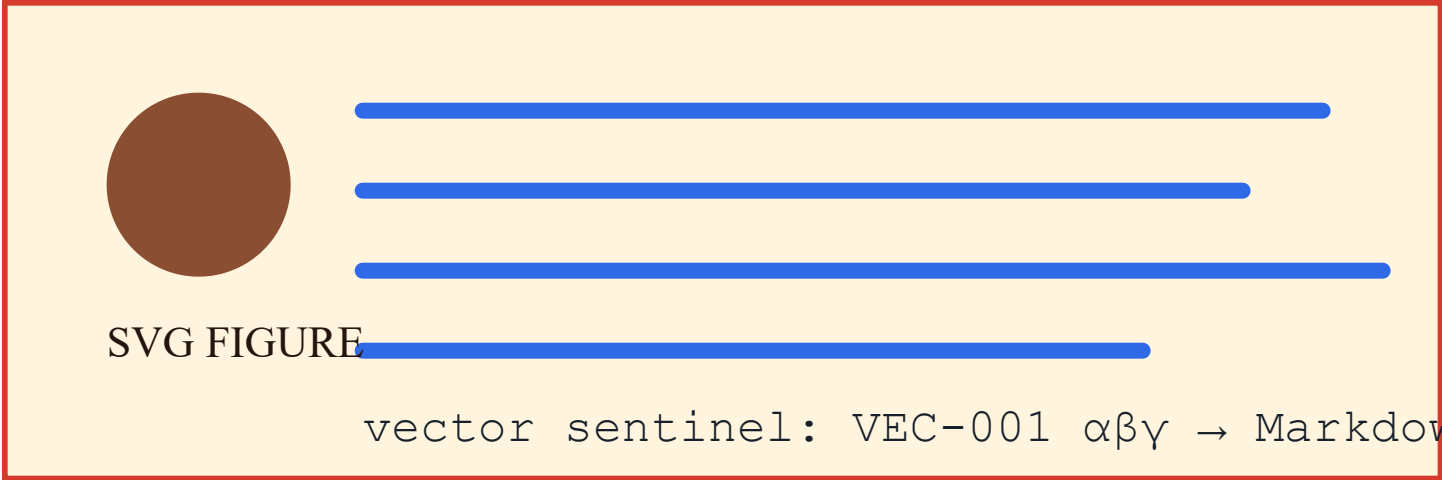
Figure 2. HTML figure with explicit figcaption and vector source.

SENTINEL-FIGURE-002 inline icon-like image should not destroy surrounding text:



after-icon.

SENTINEL-FIGURE-003 image wrapped in link:



14. OCR / Raster Hybrid Future Fixture

SENTINEL-OCR-001: The following image is intentionally raster text. V1/V2/V3 core without OCR should preserve it as an image or alt text, not hallucinate body text. V4 optional OCR may recover the raster text only when explicitly enabled and must report OCR provenance.

OCR-ONLY SENTINEL: Raster text 12345

Greek-like: alpha beta gamma; symbols: <= >= != ->

Emoji label: coffee cocoa PDF markdown

Expected OCR-visible phrase if OCR is enabled: OCR-ONLY SENTINEL: Raster text 12345.

15. Footnotes and Endnotes

SENTINEL-FOOTNOTE-001: This sentence has a numeric footnote.¹ This sentence has another note with Unicode.²

SENTINEL-FOOTNOTE-002: Multiple references to one definition may occur.¹

Endnotes

[EN1] SENTINEL-ENDNOTE-001: Endnote-style paragraph that resembles a bibliography entry but is not a citation.

16. Table of Contents, Cross-References, and Bibliography

Contents

- [Plain Text, Paragraphs, Spacing, and Line Joining](#)
- [Tables](#)
- [Figures](#)

SENTINEL-CROSSREF-001: See Figure 1, Figure 2, Table 1, Section 11, Appendix A, Equation (1), and reference [1].

References

[1] Doe, Jane. "A Synthetic PDF Benchmark." Journal of Deterministic Documents, 2026. DOI: 10.0000/cocoapdf.0001.

[2] Smith, John and Roe, Alex. *Tables Without Borders*. Example Press, 2025. <https://example.com/references/2>.

Doe, J. (2024). Author-year reference style with hanging indentation expectation. Example Journal, 12(3), 45–67.

17. Math and Formula Fallback

SENTINEL-MATH-001 inline math-like text: $E = mc^2$, $a^2 + b^2 = c^2$, $\sum_i x_i = 42$.

SENTINEL-MATH-002 display formula source:

$$\int_0^1 x^2 dx = \frac{1}{3}$$

SENTINEL-MATH-003 Unicode formula:

$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $y^2 = x$ when $x \geq 0$.

SENTINEL-MATH-004 matrix-like text:

Formula	Meaning
$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$	matrix literal
$\det(A) = -2$	determinant

18. Multi-Column, Sidebars, Callouts, and Floats via HTML/CSS

SENTINEL-COLUMNS-001 left/right flow: Column content alpha begins here. It should be read in column order when the PDF physically lays it out in columns.

Column content bravo contains Greek αβγ and numbers 12345.

Column content charlie contains a link-like visible string <https://example.com/columns>.

Column content delta ends the multi-column region.

SENTINEL-CALLOUT-001: This is an HTML callout/sidebar. Markdown cannot represent this exactly; CocoaPDF should use safe HTML fallback or report a callout region.

19. Forms and Annotations Future Fixture

SENTINEL-FORM-001 visible form-like text:

Name:

☒ checked checkbox visible label

☐ unchecked checkbox visible label

SENTINEL-ANNOTATION-001: Highlight, underline, strikeout, sticky-note, stamp, and file-attachment annotations require PDF-level fixture construction. This Markdown source includes visible labels only; annotation dictionaries should be added by a PDF fixture generator for V2.12/V3 tests.

20. Active Content and Security Future Fixture

SENTINEL-SECURITY-001: Generated PDFs for this section may be post-processed to inject `/OpenAction`, `/AA`, `/JavaScript`, `/Launch`, `/RichMedia`, remote `/GoToR`, and embedded-file actions. CocoaPDF must detect and report these without executing, fetching, launching, or submitting anything.

SENTINEL-SECURITY-002: Visible unsafe strings: `javascript:alert(1)`, `file:///tmp/secret.txt`, `data:text/html;base64,SGVsbG8=`.

21. Page Breaks, Furniture, Headers, Footers

SENTINEL-PAGE-001: The generator may insert page numbers and running headers/footers. CocoaPDF must remove repeated furniture only with repeated-page evidence and must never drop body text.

SENTINEL-PAGE-002: New page marker after explicit page break. If a PDF generator honors the break, this starts a new page.

Running header text candidate: CocoaPDF Fixture Header
Running footer text candidate: Page N Footer

22. Damaged, Encrypted, and Hard-PDF V3 Matrix Labels

SENTINEL-V3-001: This source cannot itself create malformed xrefs, object streams, encrypted streams, corrupted filters, or damaged trailers. Create derivative PDFs from this generated PDF for V3 tests:

- V3-A: object streams and xref streams
- V3-B: incremental update overriding old objects
- V3-C: empty-password RC4/AES encrypted copy
- V3-D: damaged xref requiring recovery scan
- V3-E: missing or lying `/Length`
- V3-F: no-ToUnicode symbolic font
- V3-G: CJK/RTL/vertical-writing focused pages
- V3-H: decompression bomb guard fixture

23. Final Kitchen Sink Paragraph

SENTINEL-FINAL-001: CocoaPDF final mixed paragraph contains **bold αβγ**, *italic 中文*, `code_with_pipe|and_backtick`, a link to [HTTPS](#), superscript x², subscript H₂O, emoji 🍷🚀, currency €₹฿, RTL عبرية العربية, and escaped markdown characters \ ` * _ [] () # + - . ! | .

Appendix A — Expected Comparison Notes

- V1/V2 Markdown-first output may preserve complex sections as safe HTML fallback.
- V3 hard-PDF cases require derivative PDFs created after this source PDF is generated.
- V4 OCR is optional and must be disabled by default; when enabled, OCR text must carry provenance and confidence.
- Unsafe links/actions must never remain active in HTML output.
- Structural false positives are worse than plain-text fallback.

1. Footnote definition one. It includes a URL <https://example.com/footnote>, punctuation, and continuation text. [↵](#) [↵](#)

2. Footnote with Greek αβγ, emoji 🍷, and CJK 中文. [↵](#)